

XLK Microstructure Statistical Arbitrage: New-Data Progress Report

Market-neutral XLK-vs-basket arbitrage is audited separately from XLK-only sparse-basket fair-value timing. The report emphasizes spread construction, gross/no-cost economics, transaction-cost scenarios, and last-trade execution proxies.

Top Holdings

symbol	official_weight_pct	basket_weight
NVDA	14.4920	0.1861
AAPL	12.3596	0.1587
MSFT	9.2484	0.1187
AVGO	6.0025	0.0771
MU	4.4668	0.0574
AMD	4.3026	0.0552
INTC	3.4237	0.0440
CSCO	2.6563	0.0341
PLTR	2.4089	0.0309
LRCX	2.3465	0.0301
AMAT	2.2604	0.0290
ORCL	2.1327	0.0274
TXN	1.8667	0.0240
KLAC	1.6563	0.0213
IBM	1.5886	0.0204
ADI	1.4224	0.0183
QCOM	1.3824	0.0177
ANET	1.3053	0.0168
SNDK	1.2825	0.0165
APH	1.2803	0.0164

Data Diagnostics

symbol	minutes	median_spread_bps	trade_count
AAPL	42582	2.57	68484338
ADI	31149	25.72	5539665
AMAT	33850	24.07	9720939

symbol	minutes	median_spread_bps	trade_count
AMD	40758	9.17	51260690
ANET	36432	19.43	10029069
APH	38859	13.94	10984681
AVGO	39660	9.00	48065160
CSCO	45546	2.55	21108602
IBM	36215	9.59	11042119
INTC	46062	2.85	57888289
KLAC	34736	40.86	4064568
LRCX	34908	16.07	13743943
MSFT	41853	5.21	74046055
MU	38585	12.05	50893368
NVDA	46210	1.57	281818587
ORCL	37694	7.29	40996967
PLTR	43589	4.77	71844161
QCOM	37070	11.41	15491983
TXN	35122	17.49	9173623
XLK	47695	1.07	27915985

Professor Cost/Spread Leaderboard

pair	spread_type	no_cost_midpoint	quarter_spread_cost	half_spread_taker_cost	clipped_last_trade_proxy
XLK-AAPL	price_regression_residual	5859.32	2826.00	-207.31	1315.40
XLK-PLTR	price_regression_residual	1573.04	1243.92	914.80	1486.05
XLK-ORCL	price_regression_residual	1479.50	859.25	238.99	1313.93
XLK-AMD	direct_log_price	491.96	348.19	204.43	493.45
XLK-NVDA	direct_log_price	378.03	165.82	-46.39	323.04
XLK-INTC	direct_log_price	190.19	157.19	124.19	193.28
XLK-AAPL	cum_residual_return	188.05	97.55	7.06	145.25
XLK-AVGO	cum_residual_return	139.82	32.59	-74.64	136.91
XLK-IBM	price_regression_residual	1088.77	22.80	-1043.17	454.78
XLK-MSFT	direct_log_price	172.70	3.48	-165.74	137.46
XLK-PLTR	cum_residual_return	-9.75	-89.96	-170.16	-66.28
XLK-ANET	direct_log_price	23.57	-135.88	-295.33	53.28

Sparse Bid/Ask Audit

sample	old_net_bps	new_bidask_cost_bps	new_bidask_net_bps
test	-39.56	101.14	-125.45
train	70.96	128.14	-26.14

Robust Alpha Selection

decision	economic_label	selected_strategy	train_net_bps	test_net_bps
active_rule	XLK-only timing	xlk_only_timing_mid_ridge_pos_k3_zscore_mean_e3_x0.5_h0_contra_nogate_cw1950_mh180	472.55	-17.73

Top-20 Method Diagnostics

method	selected_pairs	gate_pass_pairs	raw_test_net_bps	gate_test_net_bps	test_trades
baseline	19	0	-14985.05	0.00	2578
coint_filter	1	0	-188.99	0.00	94
combined	1	0	-428.48	0.00	74
kalman_lagged	19	0	-15984.28	0.00	3097
passive_stress	19	0	-12166.22	0.00	2794
sscore_ou_gate	19	0	-9747.57	0.00	1394

No-Trade Gate Examples

pair_or_basket	method	validation_net_bps	test_net_bps	gate_selected_flag	gate_reason
XLK-AAPL	baseline	105.40	-383.41	False	2x_cost_net<0
XLK-AAPL	kalman_lagged	-17.62	-219.11	False	validation_net<=0
XLK-AAPL	passive_stress	27.38	-134.35	False	2x_cost_net<0
XLK-AAPL	sscore_ou_gate	13.25	-272.16	False	cost_share>=0.75
XLK-ADI	baseline	-207.52	-700.34	False	validation_net<=0
XLK-ADI	kalman_lagged	-367.71	-1318.10	False	validation_net<=0
XLK-ADI	passive_stress	-100.89	-622.95	False	validation_net<=0
XLK-ADI	sscore_ou_gate	-15.85	-454.27	False	validation_net<=0
XLK-AMAT	baseline	-126.12	-2745.56	False	validation_net<=0
XLK-AMAT	kalman_lagged	-97.53	-2438.48	False	validation_net<=0

Empirical Symbol Costs

symbol	median_spread_bps	p90_spread_bps	median_halfspread_bps	median_volume
XLK	1.07	8.13	0.53	23427.00
NVDA	1.57	12.47	0.79	288420.54
CSCO	2.55	7.59	1.28	24405.28
AAPL	2.57	22.87	1.28	55090.08
INTC	2.85	9.11	1.43	145094.35
PLTR	4.77	24.46	2.39	77774.00
MSFT	5.21	35.65	2.61	40339.00
ORCL	7.29	30.10	3.65	40025.24
AVGO	9.00	35.72	4.50	33688.43
AMD	9.17	36.86	4.59	59688.51
IBM	9.59	27.99	4.80	5982.00
QCOM	11.41	31.66	5.70	12445.97
MU	12.05	38.24	6.03	52313.00
APH	13.94	45.42	6.97	11321.00
LRCX	16.07	45.14	8.04	13038.96
TXN	17.49	44.67	8.75	7933.00
ANET	19.43	51.01	9.71	10214.50
AMAT	24.07	55.05	12.03	7912.77
ADI	25.72	55.53	12.86	4123.83
KLAC	40.86	75.66	20.43	1004.00

Latency Taker Costs

symbol	latency_min	buy_taker_cost_bps_median	sell_taker_cost_bps_median
AAPL	0	1.28	1.28
AAPL	1	2.22	2.39
AAPL	2	2.50	2.67
AAPL	5	2.74	3.07
ADI	0	12.86	12.86
ADI	1	12.50	12.47
ADI	2	12.58	12.47
ADI	5	12.92	12.25
AMAT	0	12.03	12.03
AMAT	1	11.81	11.72

symbol	latency_min	buy_taker_cost_bps_median	sell_taker_cost_bps_median
AMAT	2	11.97	11.82
AMAT	5	12.32	11.64
AMD	0	4.59	4.59
AMD	1	5.69	5.70
AMD	2	5.98	5.75
AMD	5	6.11	6.11
ANET	0	9.71	9.71
ANET	1	9.87	9.93
ANET	2	10.11	10.08
ANET	5	10.37	10.26

Execution-Optimized Pair Audit

stock	spread_type	policy	val_net_bps	test_net_bps	oos_decision	final_policy
INTC	price_regression_residual	taker	3083.69	-3416.85	reject_after_oos_audit	no_trade

Loss Streamline Decision

research_path	decision	test_net_bps
market_neutral_pair_or_basket	no_trade	0.00
robust_alpha_suite_selected	no_trade	-17.73
fixed_bps_xlk_only_timing_candidate	legacy_candidate_shape_only	613.35
expanded_fixed_bps_xlk_only_timing	no_trade	-977.24
timing_robustness_current_selection	no_trade	0.00

Expanded Fixed-BPS Timing Controls

decision	basket_symbols	train_net_bps	validation_net_bps	test_net_bps	circular_pvalue
no_trade	NVDA AAPL MSFT AVGO MU	-2115.46	-45.13	-977.24	0.80

Microstructure Refinement Horizon Sweep

horizon_min	decision	train_net_bps	validation_net_bps	test_net_bps	test_trades
5	no_trade	0.00	0.00	0.00	0.00
15	no_trade	0.00	-44.37	0.00	0.00

horizon_min	decision	train_net_bps	validation_net_bps	test_net_bps	test_trades
30	no_trade	-39.94	-40.85	0.39	14.00
60	no_trade	-14.90	-213.33	-47.30	40.00

Current Timing Robustness Re-Audit

decision	reason	train_net_bps	test_net_bps
no_trade	No XLK-only timing rule passed Jan-Feb train filters.	0.00	0.00

Named Timing Candidate Re-Audit

strategy	basket_symbols	train_net_bps	mar_net_bps	apr_net_bps	test_net_bps
micro_shrink_0.75_cw10d_e60_x0_mh240	NVDA AAPL MSFT AVGO MU	-142.16	-32.45	-862.35	-894.79

Regime Shift Summary

metric	train_avg	test_avg	test_minus_train
xlk_basket_corr_1m	0.66	0.63	-0.02
xlk_on_basket_beta_1m	0.61	0.62	0.00
residual_vol_bps_1m	4.98	5.12	0.14
median_xlk_spread_bps	0.71	1.02	0.31
signal_std_bps	114.47	150.39	35.91
alpha_ic_5m	-0.01	0.03	0.04
contrarian_decile_edge_5m	-1.93	2.38	4.31
alpha_ic_15m	-0.03	0.04	0.07
contrarian_decile_edge_15m	-6.90	5.88	12.79
alpha_ic_30m	-0.06	0.05	0.11
contrarian_decile_edge_30m	-13.08	10.24	23.32
alpha_ic_60m	-0.08	0.08	0.16

Robust Alpha Controls

control	train_net_bps	test_net_bps	test_trades
selected	472.55	-17.73	10.00
sign_flip	-484.46	-23.47	10.00
active_always_long	312.15	-11.46	10.00

control	train_net_bps	test_net_bps	test_trades
active_always_short	-324.06	-29.74	10.00
circular_shift_mean		-61.70	
circular_shift_p95		42.76	
selected_vs_circular_pvalue		0.24	

XLK-Only Timing

period	gross_bps	cost_bps	net_bps	trades
train	-125.87	557.20	-683.07	120
validation	407.97	202.23	205.74	46
test	-550.25	302.18	-852.43	99
all	-268.15	1061.61	-1329.76	265





